⑯ $\int_1^{\pi}\sin\left(\frac{1}{\cos x}\right)\frac{dx}{\sqrt{x}}$ – особеност в $x=\pi/2$, понеже $\cos\frac{\pi}{2}=0$, $\frac{1}{\cos\frac{\pi}{2}}\xrightarrow[x\to\pi/2]{}\infty$

$\int_1^{\pi}=\int_1^{\pi/2}+\int_{\pi/2}^{\pi}$, $\int_1^{\pi/2}=\left(x=\frac{\pi}{2}-y\right)=-\int_{\frac{\pi}{2}-1}^{0}\sin\left(\frac{1}{\cos(\frac{\pi}{2}-y)}\right)\frac{dy}{\sqrt{\frac{\pi}{2}-y}}=\int_0^{\frac{\pi}{2}-1}\sin\left(\frac{1}{\sin y}\right)\frac{dy}{\sqrt{\frac{\pi}{2}-y}}$

Този интеграл има особеност в 0-та. Въпреки, че $\sqrt{\frac{\pi}{2}-y}\underset{y\to0}{\sim}1$, то не може да се замести $\sqrt{\frac{\pi}{2}-y}$ с 1, понеже $\sin\left(\frac{1}{\sin y}\right)$ си мени знака, т.е. интегрантът не е от неотрицателна ф-я. Не може да се използва и развитието на $\sin(\sim)$ понеже $\frac{1}{\sin y}\xrightarrow[y\to0]{}\infty$.

Има два начина за решаване – интегриране по части или използване на някой от критериите на Абел или Дирихле.

Решение чрез критерия на Дирихле:

$$\int_1^{\pi/2}\to\left(\frac{1}{\cos x}\right)'=\frac{\sin x}{\cos^2 x}\Rightarrow\int_1^{\pi/2}=\int_1^{\pi/2}\underbrace{\sin\left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos^2 x}\right)}_{f}\underbrace{\frac{\cos^2 x}{\sin x}\cdot\frac{dx}{\sqrt{x}}}_{g}$$

$$\int_1^A f(x)dx=\int_1^A\sin\left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos^2 x}\right)dx=\int_1^A\sin\left(\frac{1}{\cos x}\right)d\frac{1}{\cos x}=-\cos\left(\frac{1}{\cos x}\right)\Big|_1^A\Rightarrow$$

$\forall A<\frac{\pi}{2}\to\left|\int_1^A f(x)dx\right|<2$, т.е. $\int_1^A f$ са равномерно ограничени.

$$g'=\frac{1}{x\sin^2 x}\left[2\cos x(-\sin x)\sqrt{x}\sin x-\cos^2 x\left(\cos x.\sqrt{x}+\frac{1}{2\sqrt{x}}\sin x\right)\right]=$$

$$=-\frac{1}{x\sin^2 x}\left(2\cos x.\sin x.\sqrt{x}.\sin x+\cos^2 x.\cos x.\sqrt{x}+\cos^2 x.\frac{1}{2\sqrt{x}}\sin x\right)<0,$$ тъй като

при $x\in(1,\pi/2)\to\cos x>0,\sin x>0$. Следователно $g(x)\searrow$.

Също $\lim\limits_{x\to\pi/2}g(x)=\lim\limits_{x\to\frac{\pi}{2}}\frac{\cos^2 x}{\sin x}\cdot\frac{1}{\sqrt{x}}=0\Rightarrow$ условията в критерия на Дирихле са изпълнени $\Rightarrow\int_1^{\pi/2}$ е сходящ. Аналогично и за $\int_{\pi/2}^{\pi}$.

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⑰ $\int_0^{\infty}\frac{x\sin(5x)}{25+x^2}dx$ – особеност единствено в ∞.

$$\int_0^{\infty}\frac{x\sin(5x)}{25+x^2}dx=-\frac{1}{5}\int_0^{\infty}\frac{x\,d\cos(5x)}{25+x^2}=-\frac{1}{5}\frac{x\cos(5x)}{25+x^2}\Big|_0^{\infty}+\frac{1}{5}\int_0^{\infty}\cos(5x)\,d\frac{x}{25+x^2}$$

$$\lim_{x\to\infty}\frac{x\cos(5x)}{25+x^2}=0\Rightarrow\frac{1}{5}\frac{x\cos 5x}{25+x^2}\Big|_0^{\infty}=0\Rightarrow\int_0^{\infty}=\frac{1}{5}\int_0^{\infty}\cos(5x)\frac{(25+x^2)-2x^2}{(25+x^2)^2}dx=$$

$$=\frac{1}{5}\underbrace{\int_0^{\infty}\cos(5x)\frac{25-x^2}{(25+x^2)^2}dx}_{I}.\quad|I|=\int_0^{\infty}|\cos(5x)|\frac{|25-x^2|}{(25+x^2)^2}dx<\int_0^{\infty}\frac{x^2}{x^4}dx=\int_0^{\infty}\frac{dx}{x^2}\to\text{сх.}$$

$\Rightarrow I$ е абсолютно сходящ $\Rightarrow$ сходящ.

Може да се реши и чрез критерия на Дирихле.

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⑱ $\int_0^{\infty}\frac{\sin(x+x^2)}{x^{\alpha}}dx$ – особености в 0 и $\infty\Rightarrow\int_0^{\infty}=\int_0^1+\int_1^{\infty}$

$\int_0^1\to x+x^2\underset{x\to0}{\sim}x\Rightarrow\sin(x+x^2)\underset{x\to0}{\sim}x+x^2\sim x\Rightarrow\int_0^1\sim\int_0^1\frac{x}{x^{\alpha}}dx=\int_0^1\frac{dx}{x^{\alpha-1}}\to$ сх. при $\alpha-1<1$, т.е. $\alpha<2$.

$\int_1^{\infty}\to$ тъй като $(x+x^2)'=1+2x\to\int_1^{\infty}=\int_1^{\infty}\frac{\sin(x+x^2)(1+2x)}{(1+2x)x^{\alpha}}dx=\int_1^{\infty}\frac{\sin(x+x^2)d(x+x^2)}{(1+2x)x^{\alpha}}=$

$$=-\int_1^{\infty}\frac{d\cos(x+x^2)}{(1+2x)x^{\alpha}}=-\frac{\cos(x+x^2)}{(1+2x)x^{\alpha}}\Big|_1^{\infty}+\int_1^{\infty}\cos(x+x^2)\,d\frac{1}{(1+2x)x^{\alpha}}$$

① $\alpha>0\to\lim\limits_{x\to\infty}\frac{\cos(x+x^2)}{(1+2x)x^{\alpha}}=0\Rightarrow\int_1^{\infty}\sim\int_1^{\infty}\underbrace{\cos(x+x^2)\frac{2x^{\alpha}+\alpha x^{\alpha-1}(1+2x)}{(1+2x)^2x^{2\alpha}}}_{f(x)}dx$

$\int_1^{\infty}|f(x)|dx\le\int_1^{\infty}\frac{cx^{\alpha}}{x^{2+2\alpha}}dx\sim\int_1^{\infty}\frac{dx}{x^{2+\alpha}}\to$ сходящ

② $\alpha=0\to\lim\limits_{x\to\infty}\frac{\cos(x+x^2)}{(1+2x)x^{\alpha}}=0$, $\int_1^{\infty}\sim\int_1^{\infty}\cos(x+x^2)\,d\frac{1}{2x+1}=\int_1^{\infty}\frac{\cos(x+x^2)(-2)}{(2x+1)^2}dx$

Откъдето $\int_1^{\infty}\frac{|\cos(x+x^2)|}{(2x+1)^2}dx<\int_1^{\infty}\frac{c\,dx}{x^2}\to$ сх. $\Rightarrow\int_1^{\infty}$ е сх. при $\alpha\ge0$

③ $\alpha<0$

3.1. $-1<\alpha<0\to$ тогава $\lim\limits_{x\to\infty}\frac{\cos(x+x^2)}{(1+2x)x^{\alpha}}=0\Rightarrow\int_1^{\infty}\sim\int_1^{\infty}f(x)dx$ и

$$\left|\frac{2x^{\alpha}+\alpha x^{\alpha-1}(1+2x)}{(1+2x)^2x^{2\alpha}}\right|<\frac{c(x^{\alpha}+x^{\alpha-1})}{x^{2+2\alpha}}<\frac{cx^{\alpha}}{x^{2+2\alpha}}=\frac{c}{x^{2+\alpha}}$$ и $2+\alpha>1\Rightarrow$ сходящ

3.2. $\alpha\le-1\to\int_1^{\infty}\frac{\sin(x+x^2)}{x^{\alpha}}dx=\int_1^{\infty}x^{\beta}\sin(x+x^2)dx$. Ще покажем, че в този случай е разходящ чрез принципа на Коши. $(-\alpha=\beta\ge1)$.

$$\int_1^{\infty}x^{\beta}\sin\left(\left(x+\frac{1}{2}\right)^2-\frac{1}{4}\right)dx=\left(x+\frac{1}{2}=y\right)=\int_{3/2}^{\infty}\left(y-\frac{1}{2}\right)^{\beta}\sin\left(y^2-\frac{1}{4}\right)dy.$$

Принцип на Коши за $\int_a^{\infty}f(x)dx$.

$\forall\varepsilon>0\;\exists A\;\forall A',A''>A$ е изпълнено $\left|\int_{A'}^{A''}f(x)dx\right|<\varepsilon$.

Отрицание: $\exists\varepsilon_0\;\forall A\;\exists A',A''>A$, че $\left|\int_{A'}^{A''}f(x)dx\right|\ge\varepsilon_0$.

Правило за доказване на разходимост:

Да се намерят две редици A'_n,A''_n такива, че $A'_n\xrightarrow[n\to\infty]{}\infty$, $A''_n\xrightarrow[n\to\infty]{}\infty$ и $\left|\int_{A'_n}^{A''_n}f(x)dx\right|>\varepsilon_0$.

Нека $\left.\begin{aligned}A'_n&=\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}\\A''_n&=\sqrt{2\pi n+\frac{\pi}{2}+\frac{1}{4}}\end{aligned}\right\}\Rightarrow y^2-\frac{1}{4}\in\left[2\pi n+\frac{\pi}{4},2\pi n+\frac{\pi}{2}\right]\Rightarrow\sin\left(y^2-\frac{1}{4}\right)>\frac{\sqrt{2}}{2}\Rightarrow$

$$\left|\int_{A'_n}^{A''_n}\right|>\frac{\sqrt{2}}{2}\int_{A'_n}^{A''_n}\left(y-\frac{1}{2}\right)^{\beta}dy>\frac{\sqrt{2}}{2}\left(\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}-\frac{1}{2}\right)^{\beta}(A''_n-A'_n)>$$

$$>\frac{\sqrt{2}}{2}\left(\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}-\frac{1}{2}\right)\left(\sqrt{2\pi n+\frac{\pi}{2}+\frac{1}{4}}-\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}\right)=$$

$$=\frac{\sqrt{2}}{2}\frac{2\pi n+\pi/4}{\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}+\frac{1}{2}}\frac{\pi/4}{\sqrt{2\pi n+\frac{\pi}{2}+\frac{1}{4}}+\sqrt{2\pi n+\frac{\pi}{4}+\frac{1}{4}}}>$$

$$>\frac{\sqrt{2}}{2}\frac{2\pi n}{2\sqrt{2\pi n}}\frac{\pi/4}{2\sqrt{2\pi n+\frac{\pi}{2}+\frac{1}{4}}}>\frac{\sqrt{2}}{2}\frac{\sqrt{2\pi n}}{2}\frac{\pi/4}{2\sqrt{3\pi n}}=$$

$$=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{2\pi}}{2}\frac{\pi/4}{2\sqrt{3\pi}}>0\Rightarrow\int$$ е разходящ.

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⑲ $\int_0^{\infty}\frac{\sin\ln x}{\sqrt{x}}dx$ – особености в 0 и $\infty\Rightarrow\int_0^{\infty}=\int_0^1+\int_1^{\infty}$

$\int_0^1\to\int_0^1\frac{|\sin\ln x|}{\sqrt{x}}dx\le\int_0^1\frac{dx}{\sqrt{x}}$ – сходящ (сравнете с $\int_0^a\frac{dx}{x^{\alpha}}$)

$\int_1^{\infty}\frac{\sin\ln x}{\sqrt{x}}dx=(x=e^t)=\int_0^{\infty}\frac{\sin t}{e^{t/2}}de^t=\int_0^{\infty}e^{t/2}\sin t\,dt$ – разходящ, понеже

ако $A'_n=2\pi n+\frac{\pi}{4}$, $A''_n=2\pi n+\pi/2\Rightarrow\int_{A'_n}^{A''_n}>\int_{A'_n}^{A''_n}e^{A'_n}\cdot\frac{\sqrt{2}}{2}dt>e^{2\pi n+\frac{\pi}{4}}\cdot\frac{\sqrt{2}}{2}(A''_n-A'_n)=$

$=\frac{\sqrt{2}}{2}e^{2\pi n+\pi/4}\cdot\frac{\pi}{4}>\frac{\sqrt{2}}{2}e^{\pi/4}\cdot\pi/4>0\Rightarrow$ от принципа на Коши следва разходимост. Това е зад. 118, стр. 273, том II Кудрявцев.

Отговорът там е грешен!

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⑳ $\int_1^{\infty}\operatorname{arctg}\frac{\cos x}{x^{2/3}}dx$ – Особеност в ∞. Тъй като $\frac{\cos x}{x^{2/3}}\xrightarrow[x\to\infty]{}0$

и $\operatorname{arctg}x=x+O(x^3)\Rightarrow\operatorname{arctg}\frac{\cos x}{x^{2/3}}=\frac{\cos x}{x^{2/3}}+O\left(\frac{\cos^3 x}{x^2}\right)=\cos\frac{x}{x^2}+O\left(\frac{1}{x^2}\right)$

$\int^{\infty}O(\frac{1}{x^2})dx\to$ сх. $\Rightarrow\int_1^{\infty}\operatorname{arctg}\frac{\cos x}{x^{2/3}}dx\sim\int_1^{\infty}\frac{\cos x}{x^{2/3}}dx\to$ сх. Сравнете

с $\int^{\infty}\frac{\cos x}{x^{\alpha}}dx$, която е сх. при $\alpha>0$.

㉑ $\int_1^{\infty}\frac{1}{\sqrt[3]{x}}\left(\exp\left(\frac{\sin\sqrt{x}}{1+x}\right)-1\right)dx$ – Особеност в ∞. Тъй като

$\frac{\sin\sqrt{x}}{1+x}\xrightarrow[x\to\infty]{}0$, то $\exp\left(\frac{\sin\sqrt{x}}{1+x}\right)=1+\frac{\sin\sqrt{x}}{1+x}+O\left(\frac{1}{x^3}\right)$

Последното е изпълнено, понеже $|\sin\sqrt{x}|<1\Rightarrow\left(\frac{\sin\sqrt{x}}{1+x}\right)^3<\frac{1}{x^3}$. Тогава

$$\int_1^{\infty}=\int_1^{\infty}\frac{1}{\sqrt[3]{x}}\left(\frac{\sin\sqrt{x}}{1+x}+O\left(\frac{1}{x^3}\right)\right)dx=\int_1^{\infty}\frac{\sin\sqrt{x}}{\sqrt[3]{x}(1+x)}dx+\int_1^{\infty}O\left(\frac{1}{x^{10/3}}\right)dx$$

$\int_1^{\infty}\frac{|\sin\sqrt{x}|}{\sqrt[3]{x}(1+x)}dx<\int_1^{\infty}\frac{dx}{x^{4/3}}$ – сходящ, $\int_1^{\infty}O\left(\frac{1}{x^{10/3}}\right)dx<\int_1^{\infty}\frac{dx}{x^{10/3}}\to$ сх.

Окончателно $\int_1^{\infty}\to$ сходящ.

㉒ $\int_0^{\infty}\frac{x\cos(7x)}{x^2+2x+2}dx$ – особеност в ∞.

!Въпреки, че $x^2+2x+2\underset{x\to\infty}{\sim}x^2$ не можем да заместим x^2+2x+2 с x^2, понеже $f(x)=\frac{x\cos 7x}{x^2+2x+2}$ си мени знака.

$$\int_0^{\infty}=\frac{1}{7}\int_0^{\infty}\frac{x\,d\sin(7x)}{x^2+2x+2}=\frac{1}{7}\frac{x\sin(7x)}{x^2+2x+2}\Big|_0^{\infty}-\frac{1}{7}\int_0^{\infty}\sin(7x)\,d\frac{x}{x^2+2x+2}$$

$\lim\limits_{x\to\infty}\frac{x\sin(7x)}{x^2+2x+2}=0\Rightarrow\frac{1}{7}\frac{x\sin(7x)}{x^2+2x+2}\Big|_0^{\infty}=0\Rightarrow$

$$\int_0^{\infty}\sim\int_0^{\infty}\sin(7x)\frac{x^2+2x+2-x(2x+2)}{(x^2+2x+2)^2}dx=\int_0^{\infty}\sin(7x)\frac{-x^2+2}{(x^2+2x+2)^2}dx$$

И тъй като $\frac{-x^2+2}{(x^2+2x+2)^2}\sim\frac{1}{x^2}$ вече имаме нужния порядък

Затова $\int_0^{\infty}|\sin(7x)|\frac{|2-x^2|}{(x^2+2x+2)^2}dx<C\int_0^{\infty}\frac{dx}{x^2}$ – сходящ.

Окончателно $\int_0^{\infty}$ – сходящ. Може и с критерия на Дирихле.

㉓ $\int_1^{\infty}\sin(x\ln x)dx$ – особеност в ∞. Тъй като

$(x\ln x)'=\ln x+1$, то $\int_1^{\infty}=\int_1^{\infty}\frac{\sin(x\ln x)}{\ln x+1}(\ln x+1)dx=$

$$=\int_1^{\infty}\frac{\sin(x\ln x)}{\ln x+1}d(x\ln x)=-\int_1^{\infty}\frac{d\cos(x\ln x)}{\ln x+1}=-\frac{\cos(x\ln x)}{\ln x+1}\Big|_1^{\infty}+\int_1^{\infty}\cos(x\ln x)\,d\frac{1}{\ln x+1}$$

$\frac{\cos(x\ln x)}{\ln x+1}\xrightarrow[x\to\infty]{}0\Rightarrow\int_1^{\infty}\sim\int_1^{\infty}\cos(x\ln x)\frac{dx}{x(\ln x+1)^2}$.

Имаме нужния порядък $\frac{1}{x\ln^2 x}$ затова разглеждаме $\int|\sim|dx$

$\int_1^{\infty}\frac{|\cos(x\ln x)|}{x(\ln x+1)^2}dx<\int_1^{\infty}\frac{dx}{x\ln^2 x}$ – сходящ (Сравнете с $\int^{\infty}\frac{dx}{x^{\alpha}\ln^{\beta}x}$)

Може да се реши и чрез Дирихле.